Atoms and Ions (p13-14)

Atoms are electrically neutral (have no charge) because they contain an equal number of _____ _____ and _____ _____ . If an electron is removed or added, the atom becomes _____ and is now called an _____ .

When an electron is removed from an atom, the atom becomes a _____ _____ _____ . A positive ion is known as a _____ . When an electron is added to an atom, the atom becomes a _____ _____ _____ . A negative ion is known as an _____ .

The charge of an ion is written after the symbol of the element and in superscript for example **Na^{+}** and **O^{-2}**.

Cations

A cation is formed when an atom loses electrons. An atom will form a positive ion if its outermost electron shell is almost empty. It will lose all the electrons in the outermost shell so that only filled shells remain.

Draw a lithium atom and a lithium cation in the space below.

Almost all cations come from _____ atoms because they have _____ electrons in the outermost shell.

Anions

An anion is formed when an atom gains electrons. This will happen when the outermost electron shell is almost full. The atom will gain electrons until the outermost shell is full.

Draw a chlorine atom and a chloride ion in the space below.

All anions come from ____ - _____ elements because they have almost _____ outermost electron shells.

Naming Ions

The name of a cation (positive ion) is the same as the name of the atom. For example a sodium atom loses one electron to become a sodium ion.

Sometimes atoms can form more than one type of cation. Iron can lose two or three electrons to become Fe^{2+} or Fe^{3+}. To distinguish between the two ions, scientists add a roman numeral to the ion name to indicate the number of electrons lost, for example Fe(II) and Fe(III).

Anions (negative ions) are named differently. The name of an anion is the same as the name for the atom but ends in **–ide**. For example, a chlorine atom becomes a chloride ion.

Questions

1. What are the three parts of an atom and what is the charge on each?

2. Which part of an atom can be lost or gained to form an ion? Why do you think this part of the atom can be lost or gained?

3. In your own words, define the term ion.

4. What is the difference between a cation and an anion?

5. If a neutral atom lost two electrons, what would be the charge of the ion?

6. If an atom has three more electrons than protons, what charge would it have?

7. Use the tables in the text book to complete the information in the table below.

Element	Ion	Name of Ion	Element	Ion	Name of Ion
Lithium			Phosphorus		
Iodine			Aluminium		
Iron (II)			Sulphur		
Oxygen			Hydrogen		
Bromine			Nitrogen		
Potassium			Sodium		
Beryllium			Fluorine		
Chlorine			Magnesium		
Copper (II)			Iron (III)		

8. Use your periodic table to predict the ions that are formed from the following atoms and give the name of each ion.

Element	Ion	Name of Ion	Element	Ion	Name of Ion
Calcium			Selenium		
Arsenic			Caesium		
Boron			Barium		
Rubidium			Astatine		

Polyatomic Ions

All the ions looked at so far have been made of one atom with a charge. These ions are called monatomic ions. Some ions are made from molecules (two or more atoms chemically combined) which have a charge. These are called polyatomic ions.

Some examples of common polyatomic ions are given below.

Ammonium	NH_4^+
Nitrate	NO_3^-
Hydroxide	OH^-
Sulphate	SO_4^{2-}
Carbonate	CO_3^{2-}
Phosphate	PO_4^{3-}